

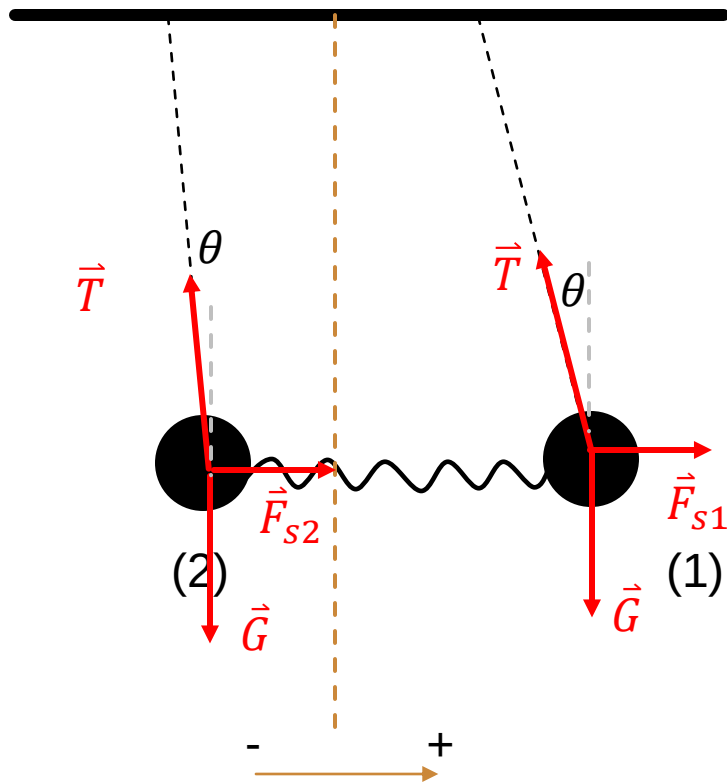


OSCILLATIONS, WAVES AND OPTICS

PHYSICS FOR CHEMISTRY STUDENTS

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REVIEW



$$m\ddot{x}_1 = -\left(k + \frac{mg}{l}\right)x_1 + kx_2$$

$$m\ddot{x}_2 = kx_1 - \left(k + \frac{mg}{l}\right)x_2$$

Coupling term

REVIEW

$$\hat{x}: \begin{cases} \ddot{x}_1 = -\left(\frac{k}{m} + \frac{g}{l}\right)x_1 + \frac{k}{m}x_2 \\ \ddot{x}_2 = \frac{k}{m}x_1 - \left(\frac{k}{m} + \frac{g}{l}\right)x_2 \end{cases}$$

$$\begin{vmatrix} \frac{k}{m} + \frac{g}{l} - \omega^2 & -\frac{k}{m} \\ -\frac{k}{m} & \frac{k}{m} + \frac{g}{l} - \omega^2 \end{vmatrix} = 0$$

$$\ddot{\mathbf{X}} = -\mathbf{M}^{-1}\mathbf{K}\mathbf{X} \Rightarrow \ddot{\mathbf{X}} = -\mathbf{\Lambda}\mathbf{X}$$

$$\mathbf{X}(t) = \mathbf{A}\cos(\omega t + \phi)$$

$$\det(\mathbf{\Lambda} - \omega^2\mathbf{I}) = 0$$

$$\mathbf{\Lambda}\mathbf{A} = \omega^2\mathbf{A} \Rightarrow (\mathbf{\Lambda} - \omega^2\mathbf{I})\mathbf{A} = \mathbf{0}$$

$\mathbf{\Lambda}$: linear transformation matrix

ω : eigenvalue

$\mathbf{A}$: eigenvector

Different ω associated with linearly independent $\mathbf{A}$

REVIEW

$$\omega_2^2 = \frac{2k}{m} + \frac{g}{l} \quad \omega_1^2 = \frac{g}{l}$$



$$\ddot{\mathbf{X}} = -\mathbf{\Lambda}\mathbf{X}$$

$$\mathbf{X}(t) = \mathbf{A}\cos(\omega t + \phi)$$



$$\mathbf{A}_1 = \alpha \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\mathbf{A}_2 = \beta \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$



General solutions:

$$\begin{aligned} \mathbf{X}_a &= \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ 1 \end{pmatrix} \cos\left(\sqrt{\frac{g}{l}} t + \phi\right) \\ \mathbf{X}_b &= \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \beta \begin{pmatrix} 1 \\ -1 \end{pmatrix} \cos\left(\sqrt{\frac{2k}{m} + \frac{g}{l}} t + \phi\right) \end{aligned}$$

$$\begin{vmatrix} \frac{k}{m} + \frac{g}{l} - \omega^2 & -\frac{k}{m} \\ -\frac{k}{m} & \frac{k}{m} + \frac{g}{l} - \omega^2 \end{vmatrix} = 0$$



$$\det(\mathbf{\Lambda} - \omega^2 \mathbf{I}) = 0$$

REVIEW

Normal Mode: A normal mode of an oscillating system is a pattern of motion in which all parts of the system move sinusoidally with the **same frequency** and **with a fixed phase relation**.

Normal modes:

$$\mathbf{X}_a = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ 1 \end{pmatrix} \cos\left(\sqrt{\frac{g}{l}} t + \phi\right)$$

$$\mathbf{X}_b = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \beta \begin{pmatrix} 1 \\ -1 \end{pmatrix} \cos\left(\sqrt{\frac{2k}{m} + \frac{g}{l}} t + \phi\right)$$

$$\mathbf{\Lambda} \mathbf{A} = \omega^2 \mathbf{A}$$

if $\mathbf{\Lambda}$ is a symmetric matrix

$\mathbf{X}_i$ mutual orthogonality of eigenvectors

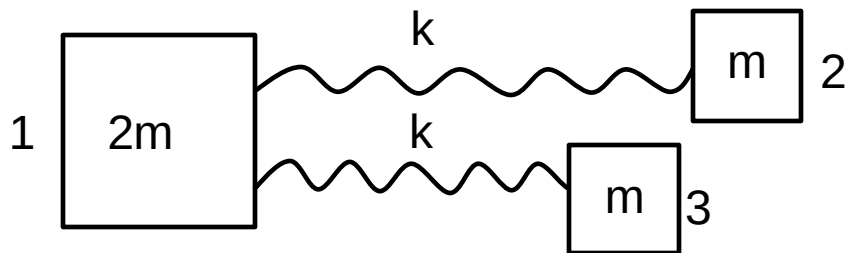
REVIEW

of the normal modes for a coupled oscillating system

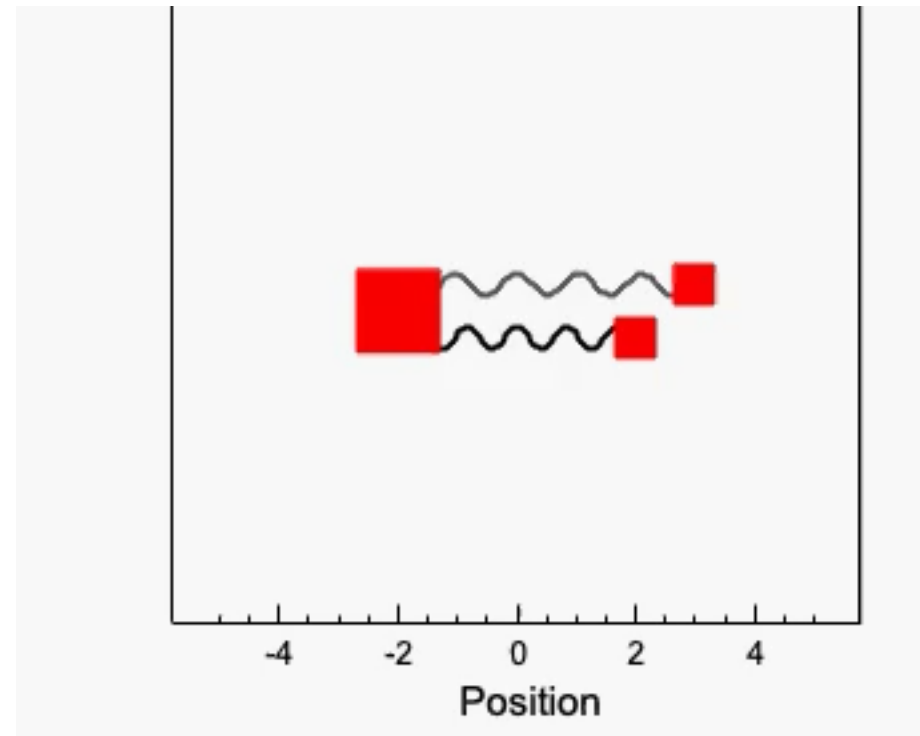


of linearly independent motion equations

2.2. NORMAL MODES



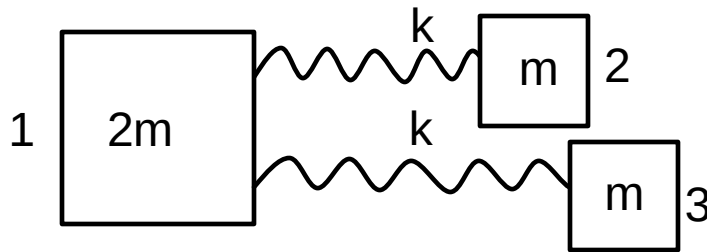
$t = 0$



2.2. NORMAL MODES

Solve the problem using mathematical techniques ...

2.2. NORMAL MODES



$$\begin{cases} 2m\ddot{x}_1 = -k[x_1 - x_2] - k[x_1 - x_3] \\ m\ddot{x}_2 = -k[x_2 - x_1] \\ m\ddot{x}_3 = -k[x_3 - x_1] \end{cases}$$

$$\ddot{\mathbf{X}} = \begin{pmatrix} \ddot{x}_1 \\ \ddot{x}_2 \\ \ddot{x}_3 \end{pmatrix} \quad \mathbf{X} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

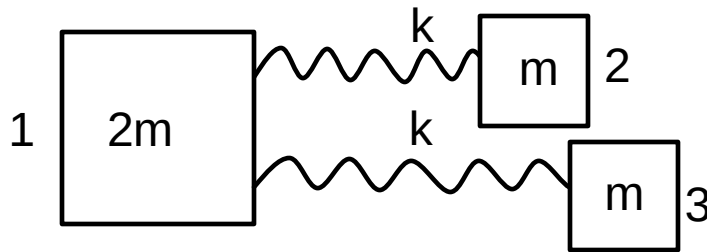
$$\begin{cases} \ddot{x}_1 = -k/m x_1 + k/2m x_2 + k/2m x_3 \\ \ddot{x}_2 = k/m x_1 - k/m x_2 \\ \ddot{x}_3 = k/m x_1 - k/m x_3 \end{cases}$$

$$\mathbf{\Lambda} = \begin{pmatrix} 2k/2m & -k/2m & -k/2m \\ -k/m & k/m & 0 \\ -k/m & 0 & k/m \end{pmatrix}$$

$$\ddot{\mathbf{X}} = -\mathbf{\Lambda}\mathbf{X}$$

2.2. NORMAL MODES

$$\ddot{\mathbf{X}} = -\Lambda \mathbf{X}$$



Trial solution : $\mathbf{X}(t) = A \cos(\omega t + \phi)$

Characteristic eq.:

$$\omega^2 \mathbf{A} = \Lambda \mathbf{A}$$

$$\det(\Lambda - \omega^2 \mathbf{I}) = 0$$

$$\det \begin{pmatrix} 2k/2m - \omega^2 & -k/2m & -k/2m \\ -k/m & k/m - \omega^2 & 0 \\ -k/m & 0 & k/m - \omega^2 \end{pmatrix} = 0$$

2.2. NORMAL MODES

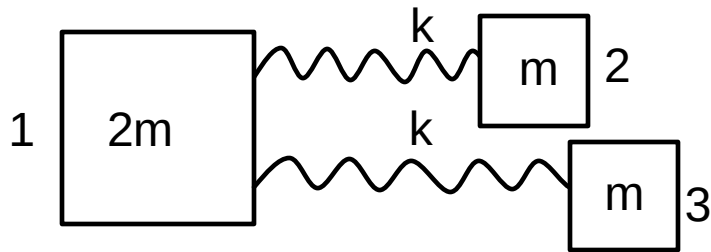
$$\det \begin{pmatrix} \frac{2k}{2m} - \omega^2 & -\frac{k}{2m} & -\frac{k}{2m} \\ -\frac{k}{m} & \frac{k}{m} - \omega^2 & 0 \\ -\frac{k}{m} & 0 & \frac{k}{m} - \omega^2 \end{pmatrix} = 0$$

$$\frac{k}{m} = \omega_0^2 \quad \Rightarrow \quad \det \begin{pmatrix} \omega_0^2 - \omega^2 & -\frac{\omega_0^2}{2} & -\frac{\omega_0^2}{2} \\ -\omega_0^2 & \omega_0^2 - \omega^2 & 0 \\ -\omega_0^2 & 0 & \omega_0^2 - \omega^2 \end{pmatrix} = 0$$

$$\Rightarrow (\omega^2 - \omega_0^2)^3 - \frac{1}{2} \omega_0^4 (\omega^2 - \omega_0^2) - \frac{1}{2} \omega_0^4 (\omega^2 - \omega_0^2) = 0$$

$$\Rightarrow \omega^2 = \omega_0^2, 2\omega_0^2, 0$$

2.2. NORMAL MODES



$$\omega^2 = 0 \quad (\Lambda - \omega^2 I)A = 0$$

$$\begin{pmatrix} 1 & -1/2 & -1/2 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$x_1(t) = x_2(t) = x_3(t) = vt + c$$

$$\ddot{X} = -\Lambda X$$

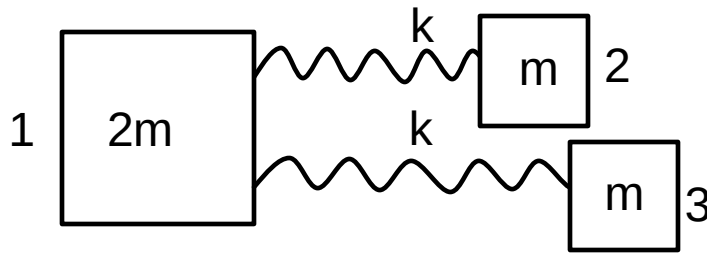
$$\omega^2 = 0$$

$$X(t) = A \cos(\phi)$$

$$\ddot{X} = -\Lambda A \cos(\phi) = 0$$

the second derivative equals zero

2.2. NORMAL MODES



$$(\Lambda - \omega^2 I)A = 0$$

$$\begin{pmatrix} \omega_0^2 - \omega^2 & -1/2 \omega_0^2 & -1/2 \omega_0^2 \\ -\omega_0^2 & \omega_0^2 - \omega^2 & 0 \\ -\omega_0^2 & 0 & \omega_0^2 - \omega^2 \end{pmatrix} A = 0$$

$$\omega^2 = \omega_0^2$$

$$\begin{pmatrix} 0 & -1/2 & -1/2 \\ -1 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix} A = 0$$

$$A = a \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

$$\omega^2 = 2\omega_0^2$$

$$\begin{pmatrix} 1 & 1/2 & 1/2 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} A = 0$$

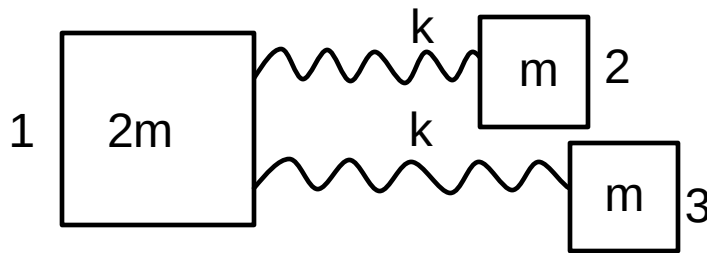
$$b \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$$

$$\omega^2 = 0$$

$$\begin{pmatrix} 1 & -1/2 & -1/2 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix} A = 0$$

$$c \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

2.2. NORMAL MODES



$$(\Lambda - \omega^2 I)A = 0$$

$$\begin{pmatrix} \omega_0^2 - \omega^2 & -1/2 \omega_0^2 & -1/2 \omega_0^2 \\ -\omega_0^2 & \omega_0^2 - \omega^2 & 0 \\ -\omega_0^2 & 0 & \omega_0^2 - \omega^2 \end{pmatrix} A = 0$$

A

$$\omega^2 = \omega_0^2$$

$$a \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

$$\omega^2 = 2\omega_0^2$$

$$b \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$$

$$\omega^2 = 0$$

$$c \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$x_1(t) = 0$$

$$x_2(t) = -x_3(t)$$

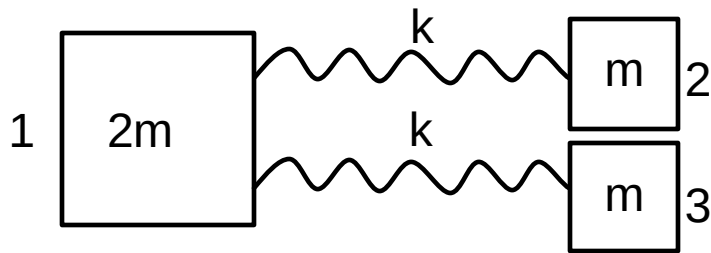
$$= a_0 \cos \left(\sqrt{\frac{k}{m}} t + \phi \right)$$

$$-x_1(t) = x_2(t) = x_3(t)$$

$$= b_0 \cos \left(\sqrt{\frac{2k}{m}} t + \phi \right)$$

$$x_1(t) = x_2(t) = x_3(t) \\ = vt + c$$

2.2. NORMAL MODES

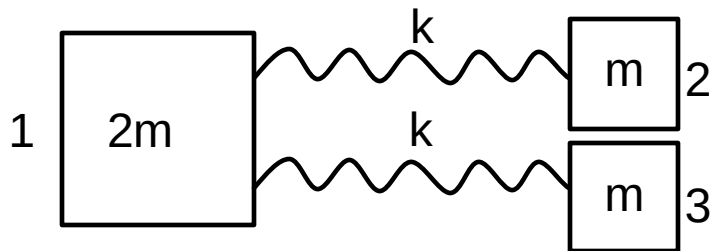


The most general motion of a system is a **superposition** of its normal modes.

Normal Modes: same frequency
fixed phase relation

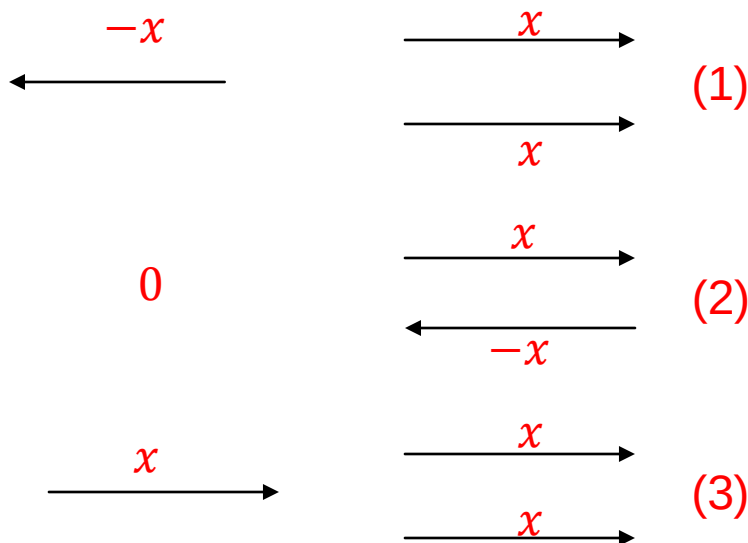
Solve the problem using physical techniques ...

2.2. NORMAL MODES

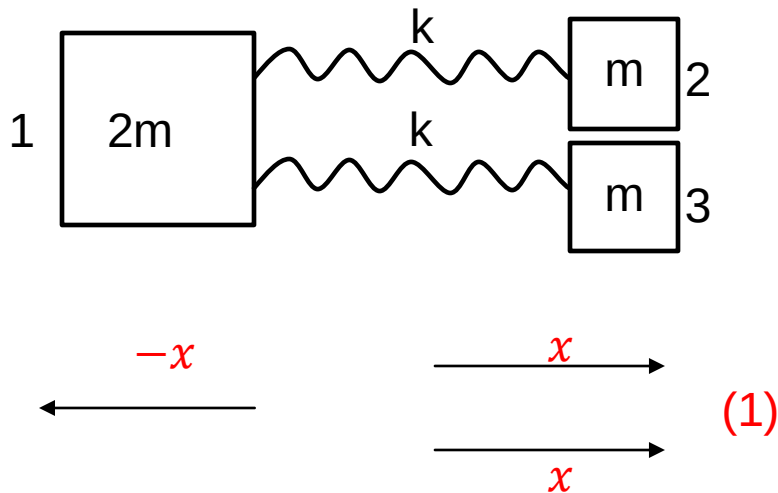


The most general motion of a system is a **superposition** of its normal modes.

Normal Modes: same frequency
fixed phase relation



2.2. NORMAL MODES



$$m\ddot{x}_2 = -k[x_2 - x_1]$$

$$\ddot{x}_2 = -2k/m \cdot x_2$$

$$m\ddot{x}_3 = -k[x_3 - x_1]$$

$$2m\ddot{x}_1 = -k[x_1 - x_2] - k[x_1 - x_3]$$

$$x_1 = -x_2$$

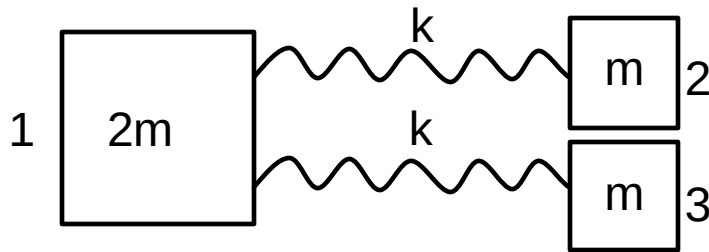
$$x_1 = -x_3$$

$$\ddot{x}_1 = -2k/m \cdot x_1$$

$$x_1(t) = a_0 \cos \left(\sqrt{\frac{2k}{m}} t + \phi \right)$$

$$x_1(t) = -x_2(t) = -x_3(t)$$

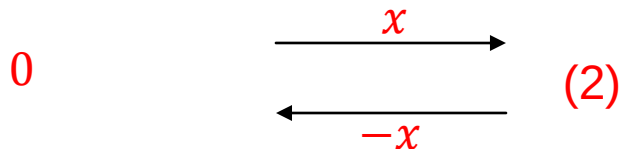
2.2. NORMAL MODES



$$m\ddot{x}_2 = -k \cdot x_2$$

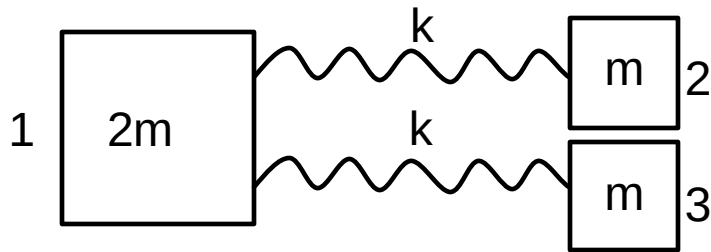
$$x_2(t) = b_0 \cos \left(\sqrt{\frac{k}{m}} t + \phi \right)$$

$$x_2(t) = -x_3(t) \quad x_1(t) = 0$$



$$x_1(t) = 0 \cdot \cos \left(\sqrt{\frac{k}{m}} t + \phi \right)$$

2.2. NORMAL MODES

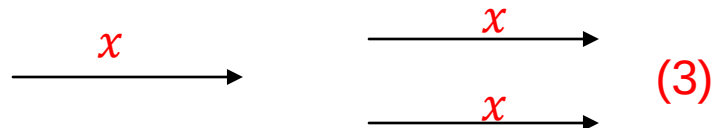


$$m\ddot{x}_1 = 0$$

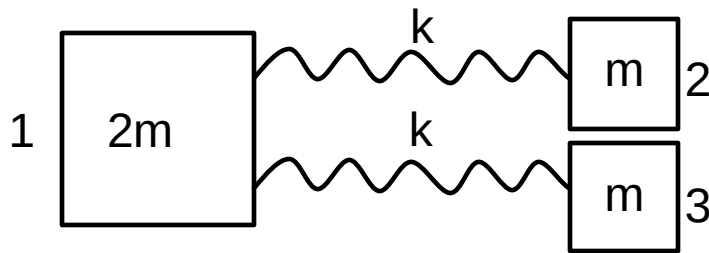
$$m\ddot{x}_2 = 0$$

$$m\ddot{x}_3 = 0$$

$$x_1(t) = x_2(t) = x_3(t) = vt + c$$



2.2. NORMAL MODES



$$m\ddot{x} = -kx(t)$$

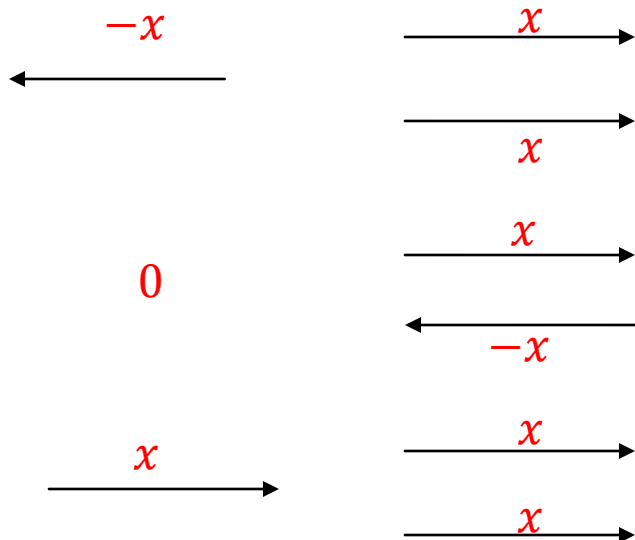
Normal Modes:

$$x_1(t) = -x_2(t) = -x_3(t) = a_0 \cos\left(\sqrt{\frac{2k}{m}}t + \phi\right)$$

(1)

(2)

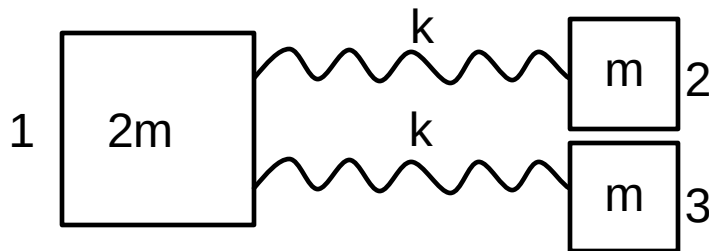
(3)



$$\begin{cases} x_1(t) = 0 \cdot \cos\left(\sqrt{\frac{k}{m}}t + \phi\right) \\ x_2(t) = -x_3(t) = b_0 \cos\left(\sqrt{\frac{k}{m}}t + \phi\right) \end{cases}$$

$$x_1(t) = x_2(t) = x_3(t) = vt + c$$

2.2. NORMAL MODES



The most general motion of a system is a **superposition** of its normal modes.

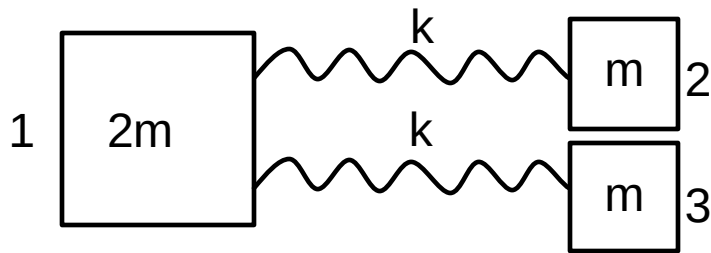
Normal Modes: same frequency
fixed phase relation

$$x_1(t) = a_0 \cos\left(\sqrt{\frac{2k}{m}}t + \phi\right) + vt + c$$

$$x_2(t) = -a_0 \cos\left(\sqrt{\frac{2k}{m}}t + \phi\right) + b_0 \cos\left(\sqrt{\frac{k}{m}}t + \phi\right) + vt + c$$

$$x_3(t) = -a_0 \cos\left(\sqrt{\frac{2k}{m}}t + \phi\right) - b_0 \cos\left(\sqrt{\frac{k}{m}}t + \phi\right) + vt + c$$

2.2. NORMAL MODES



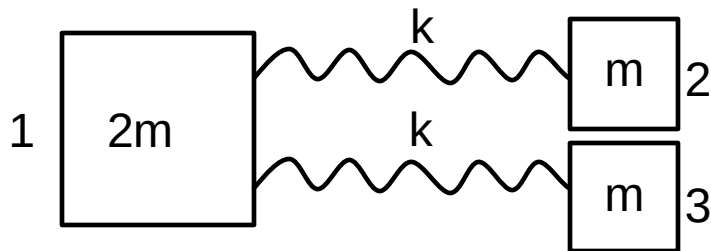
The most general motion of a system is a **superposition** of its normal modes.

Normal Modes: same frequency
fixed phase relation

Solve the problem using physical technique...

Normal Mode Frequencies : $\omega = \sqrt{\frac{2k}{m}}, \sqrt{\frac{k}{m}}, 0$

2.2. NORMAL MODES



The most general motion of a system is a **superposition** of its normal modes.

Normal Modes: same frequency
fixed phase relation

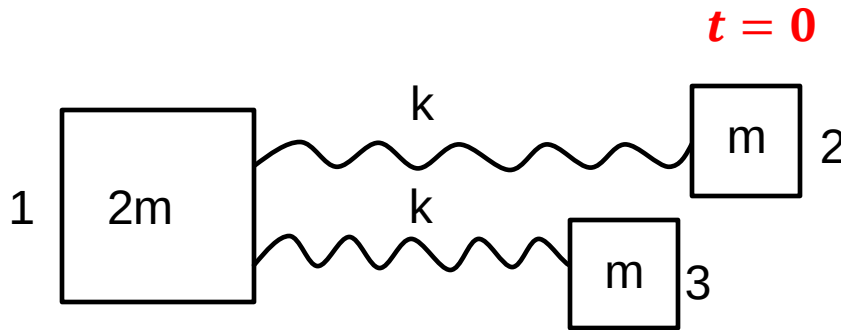
Solve the problem using physical technique...

Normal Modes: $\omega = \sqrt{\frac{2k}{m}}, \sqrt{\frac{k}{m}}, 0$

Solve the problem using mathematical techniques ...

Normal Modes: $\omega = \sqrt{\frac{2k}{m}}, \sqrt{\frac{k}{m}}, 0$

2.2. NORMAL MODES



$$\left. \begin{aligned} \dot{x}_1(0) &= 0 \\ \dot{x}_2(0) &= 0 \\ \dot{x}_3(0) &= 0 \end{aligned} \right\}$$

General solution

$$x_1(t) = a_0 \cos\left(\sqrt{\frac{2k}{m}}t + \phi\right) + vt + c$$

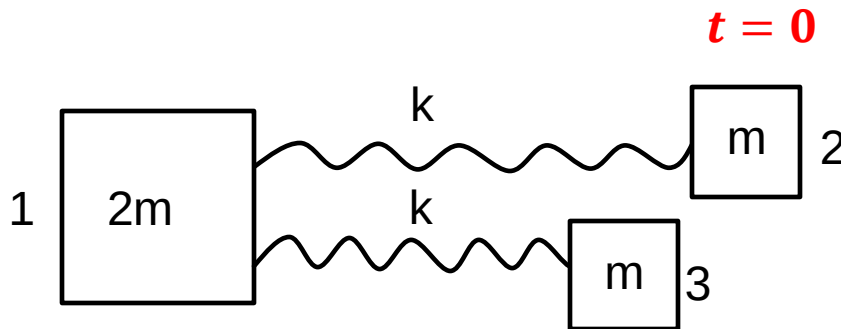
$$x_2(t) = -a_0 \cos\left(\sqrt{\frac{2k}{m}}t + \phi\right) + b_0 \cos\left(\sqrt{\frac{k}{m}}t + \phi\right) + vt + c$$

$$x_3(t) = -a_0 \cos\left(\sqrt{\frac{2k}{m}}t + \phi\right) - b_0 \cos\left(\sqrt{\frac{k}{m}}t + \phi\right) + vt + c$$



$$\begin{aligned} \sin(\phi) &= 0 \\ v &= 0 \end{aligned}$$

2.2. NORMAL MODES



$$\left. \begin{aligned} x_1(0) &= 0 \\ x_2(0) &= \alpha \\ x_3(0) &= 0 \end{aligned} \right\}$$

General solution

$$x_1(t) = a_0 \cos\left(\sqrt{\frac{2k}{m}} t\right) + c$$

$$x_2(t) = -a_0 \cos\left(\sqrt{\frac{2k}{m}} t\right) + b_0 \cos\left(\sqrt{\frac{k}{m}} t\right) + c$$

$$x_3(t) = -a_0 \cos\left(\sqrt{\frac{2k}{m}} t\right) - b_0 \cos\left(\sqrt{\frac{k}{m}} t\right) + c$$

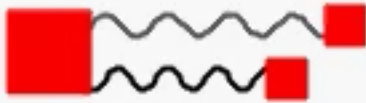


$$\begin{aligned} a_0 + c &= 0 \\ -a_0 + b_0 + c &= \alpha \\ -a_0 - b_0 + c &= 0 \end{aligned}$$



$$\begin{aligned} a_0 &= -\alpha/4 \\ b_0 &= \alpha/2 \\ c &= \alpha/4 \end{aligned}$$

2.2. NORMAL MODES



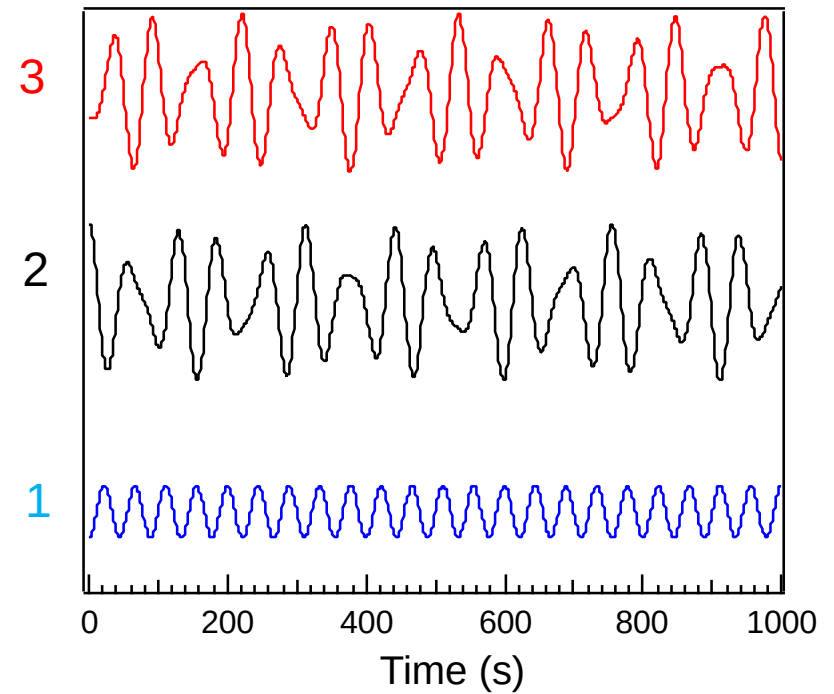
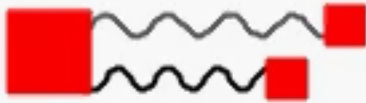
Let $\alpha = 1$

$$x_1(t) = -0.25 * \cos\left(\sqrt{\frac{k}{m}} t\right) + 0.25$$

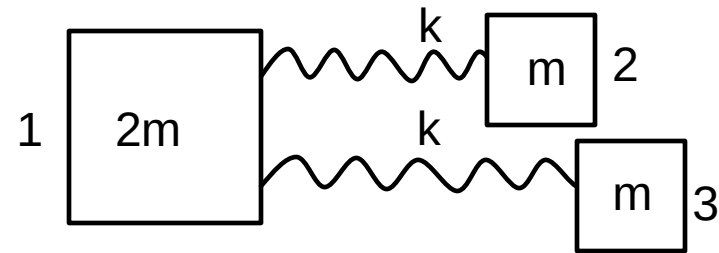
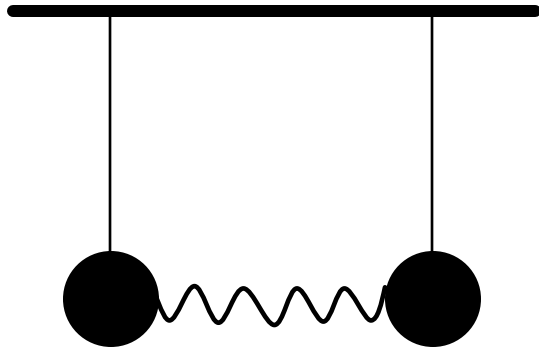
$$x_2(t) = 0.5 * \cos\left(\sqrt{\frac{2k}{m}} t\right) + 0.25 * \cos\left(\sqrt{\frac{k}{m}} t\right) + 0.25$$

$$x_3(t) = -0.5 * \cos\left(\sqrt{\frac{2k}{m}} t\right) + 0.25 * \cos\left(\sqrt{\frac{k}{m}} t\right) + 0.25$$

2.2. NORMAL MODES



2.2. NORMAL MODES



Arbitrary pump

Norm Mode pump

Motion

not harmonic

harmonic

Amplitude ratio

varies

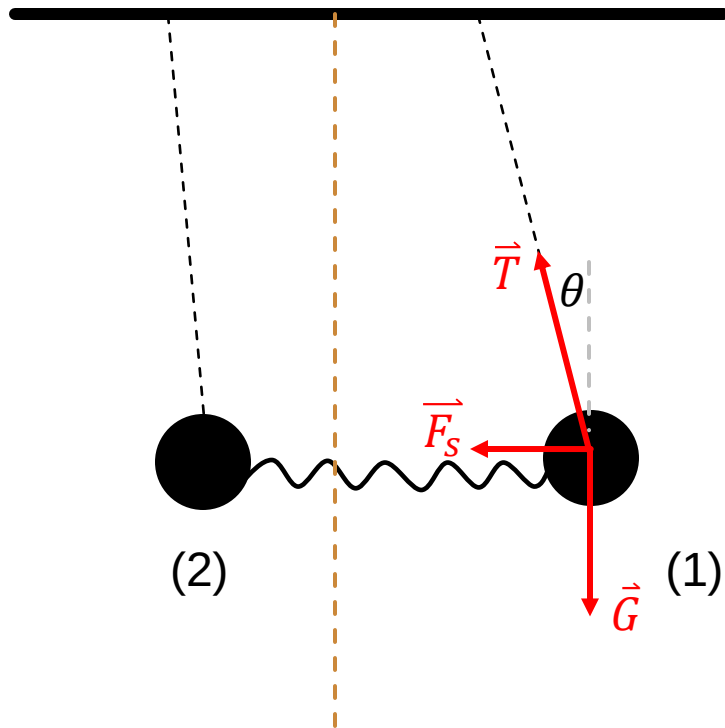
fixed

Energy

flows

stays

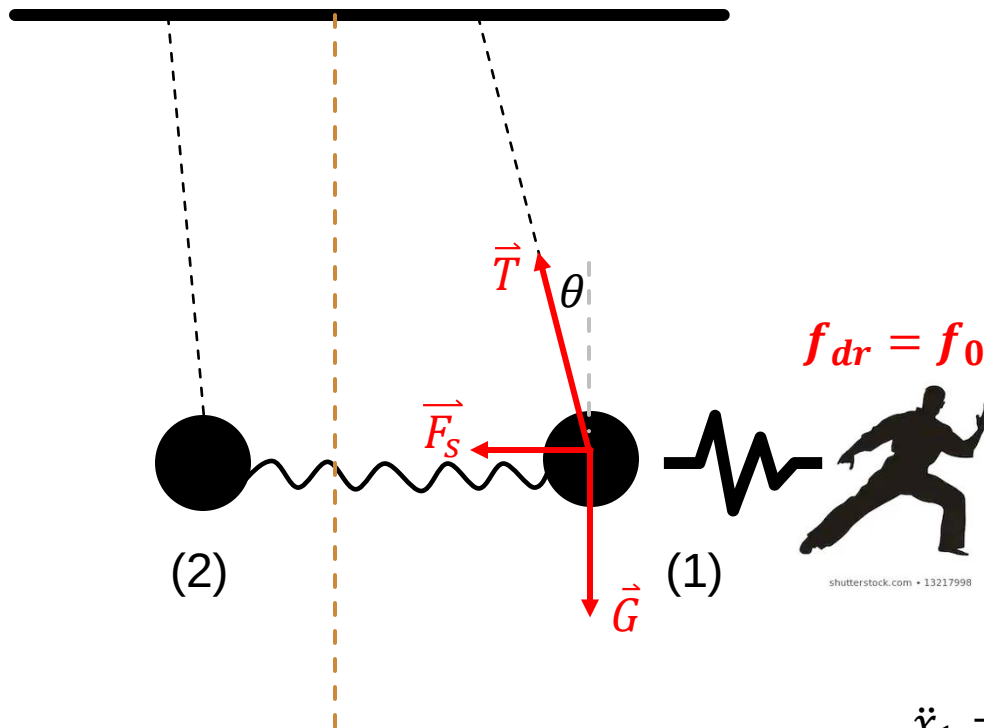
2.2 NORMAL MODES



Solve the problem using physical technique...



2.3 MOLECULAR VIBRATION



Equation of motion:

$$m\ddot{x}_1 = -\left(k + \frac{mg}{l}\right)x_1 + kx_2$$

$$m\ddot{x}_2 = kx_1 - \left(k + \frac{mg}{l}\right)x_2$$

$$f_{dr} = f_0 \cos(\omega_d t)$$

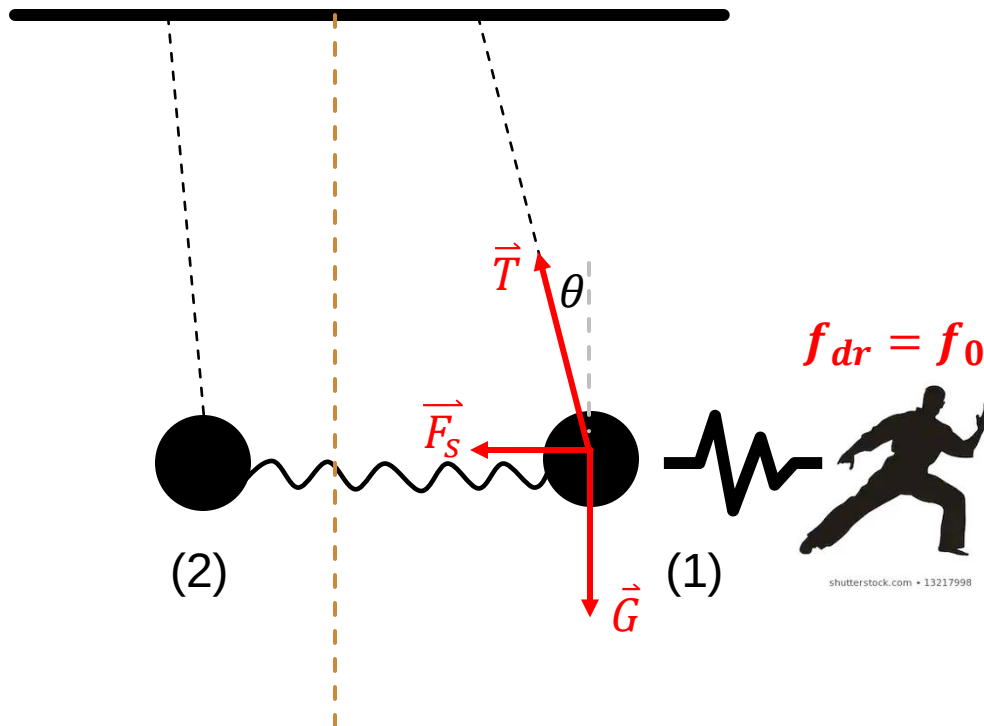
Driving force

New equation of motion:

$$\ddot{x}_1 = -\left(\frac{k}{m} + \frac{g}{l}\right)x_1 + \frac{k}{m}x_2 + \frac{f_0}{m}\cos(\omega_d t)$$

$$\ddot{x}_2 = \frac{k}{m}x_1 - \left(\frac{k}{m} + \frac{mg}{l}\right)x_2$$

2.3 MOLECULAR VIBRATION



New equation of motion:

$$\ddot{\mathbf{X}} = -\mathbf{\Lambda}\mathbf{X} + \mathbf{F}\cos(\omega_d t)$$

$$\mathbf{F} = \begin{pmatrix} f_0/m \\ 0 \end{pmatrix}$$

$$f_{dr} = f_0 \cos(\omega_d t)$$

Particular solution

$$\mathbf{X}(t) = \mathbf{B}e^{i\omega_d t}$$

$$(\mathbf{\Lambda} - \omega_d^2 \mathbf{I}) \mathbf{B} = \mathbf{F}$$

General solution of the homogeneous Eq.

$$\omega_2^2 = \frac{2k}{m} + \frac{g}{l}$$

$$\omega_1^2 = \frac{g}{l}$$

Normal modes

2.3 MOLECULAR VIBRATION

Particular solution

$$\mathbf{X}(t) = \mathbf{B}e^{i\omega_d t}$$

$$\omega_d^2 \neq \omega_1^2$$

$$\omega_d^2 \neq \omega_2^2$$

$$\begin{pmatrix} \frac{k}{m} + \frac{g}{l} - \omega_d^2 & -\frac{k}{m} \\ -\frac{k}{m} & \frac{k}{m} + \frac{g}{l} - \omega_d^2 \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} \frac{f_0}{m} \\ 0 \end{pmatrix}$$

$$(\mathbf{\Lambda} - \omega_d^2 \mathbf{I}) \mathbf{B} = \mathbf{F}$$

$$\left(\frac{k}{m} + \frac{g}{l} - \omega_d^2 \right) \cdot b_1 - \frac{k}{m} \cdot b_2 = \frac{f_0}{m}$$

$$-\frac{k}{m} \cdot b_1 + \left(\frac{k}{m} + \frac{g}{l} - \omega_d^2 \right) \cdot b_2 = 0$$

$$b_2 = \frac{\frac{k}{m}}{\left(\frac{k}{m} + \frac{g}{l} - \omega_d^2 \right)} b_1$$

$$b_1 = \frac{\frac{f_0}{m} \left(\frac{k}{m} + \frac{g}{l} - \omega_d^2 \right)}{\left(\frac{k}{m} + \frac{g}{l} - \omega_d^2 \right)^2 - \left(\frac{k}{m} \right)^2}$$

$$b_1 = \frac{\frac{f_0}{m} \left(\frac{k}{m} + \frac{g}{l} - \omega_d^2 \right)}{(\omega_1^2 - \omega_d^2) \cdot (\omega_2^2 - \omega_d^2)}$$

$$b_2 = \frac{\frac{f_0}{m} \cdot \frac{k}{m}}{(\omega_d^2 - \omega_1^2) \cdot (\omega_d^2 - \omega_2^2)}$$

2.3 MOLECULAR VIBRATION

$$b_2 = \frac{\frac{k}{m}}{(\frac{k}{m} + \frac{g}{l} - \omega_d^2)} b_1$$

$$\omega_2^2 = \frac{2k}{m} + \frac{g}{l} \quad \omega_1^2 = \frac{g}{l}$$

$$\omega_d^2 \rightarrow \omega_1^2 \quad \frac{b_1}{b_2} \rightarrow 1 \quad \Rightarrow \quad \text{System will oscillate in normal mode 1}$$

$$\omega_d^2 \rightarrow \omega_2^2 \quad \frac{b_1}{b_2} \rightarrow -1 \quad \Rightarrow \quad \text{System will oscillate in normal mode 2}$$

General solution

$$x_j(t) = x_{jc}(t) + b_j e^{i\omega_d t}$$

die out if consider damping

2.3 MOLECULAR VIBRATION

Particular solution

$$\mathbf{X}(t) = \mathbf{B}t e^{i\omega_d t}$$

$$\omega_d^2 = \omega_1^2$$

$$\omega_d^2 = \omega_2^2$$

$$\ddot{\mathbf{X}} = -\mathbf{\Lambda}\mathbf{X} + \mathbf{F}e^{i\omega_d t}$$

$$-i2\omega_d \mathbf{B} - \omega_d^2 t \mathbf{B} = -\mathbf{\Lambda} t \mathbf{B} + \mathbf{F}$$

$$x_1(t) = x_{jc}(t) + a't \sin(\omega_1 t)$$

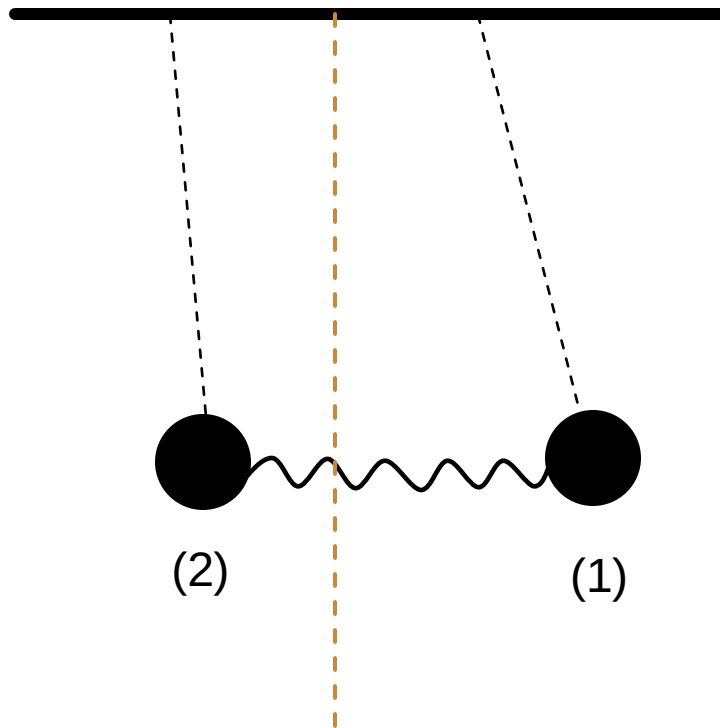
$$x_2(t) = x_{jc}(t) + a't \sin(\omega_1 t)$$



$$x_1(t) = x_{jc}(t) + b't \sin(\omega_2 t)$$

$$x_2(t) = x_{jc}(t) - b't \sin(\omega_2 t)$$

2.3 MOLECULAR VIBRATION



$$\omega_d^2 \rightarrow \omega_1^2$$

$$\frac{b_1}{b_2} \rightarrow 1$$

$$\omega_d^2 \rightarrow \omega_2^2$$

$$\frac{b_1}{b_2} \rightarrow -1$$

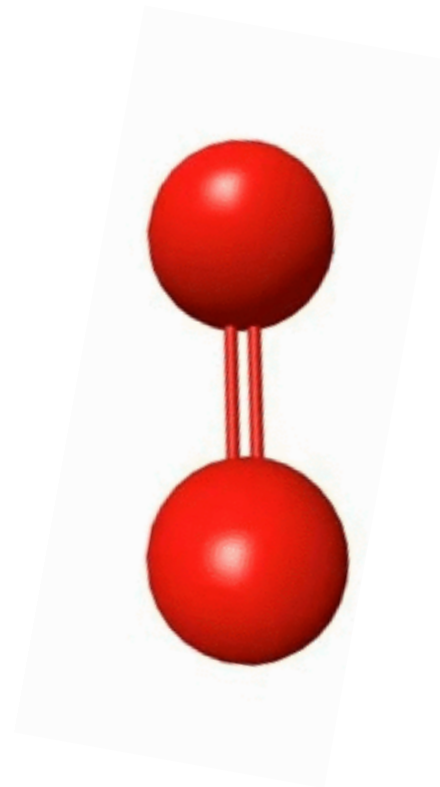
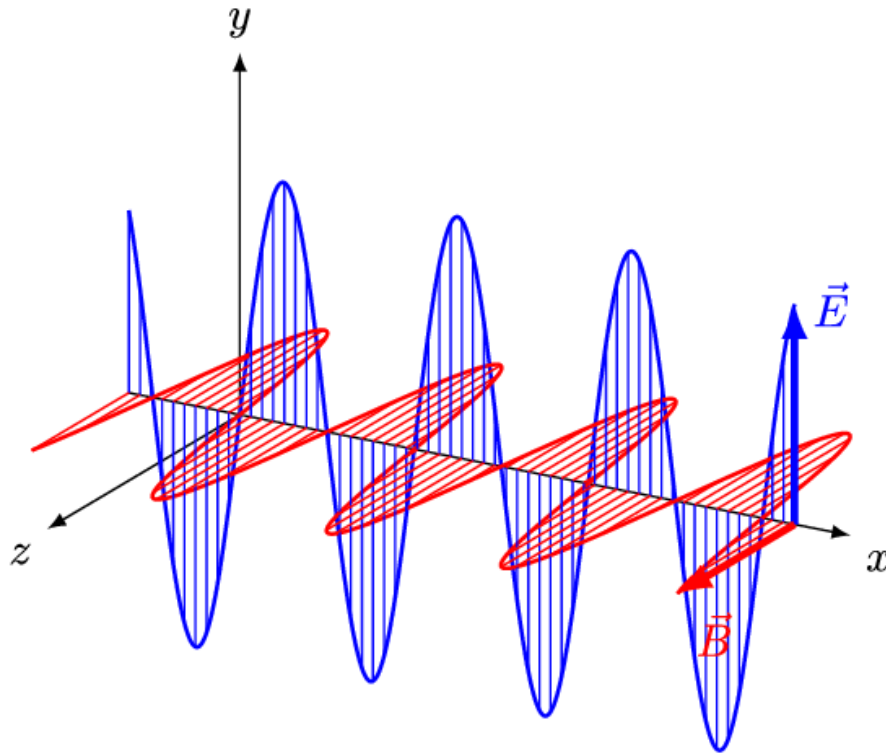
$$f_{dr} = f_0 \cos(\omega_d t)$$



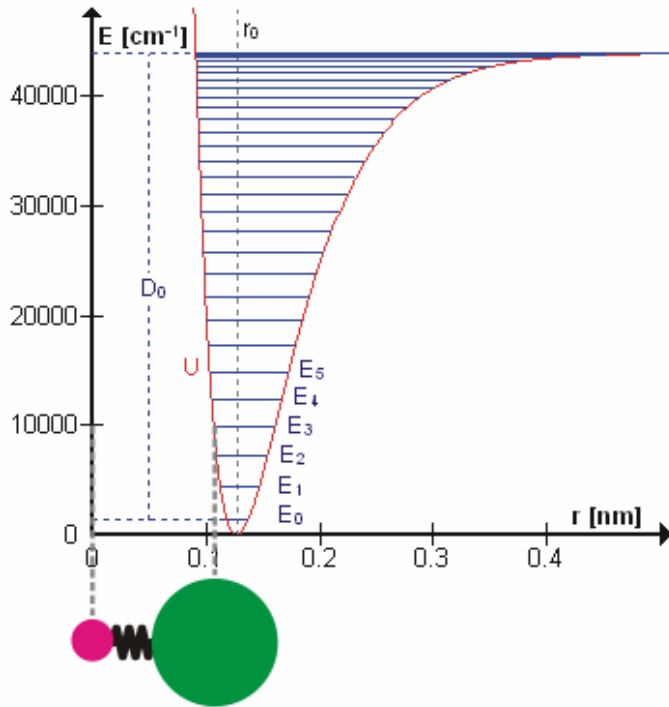
$$Q = \frac{\omega_0}{\gamma}$$

If there is damping, then Q is no longer infinite.

2.3 MOLECULAR VIBRATION



2.3 MOLECULAR VIBRATION



Diatomic molecules

one degree of freedom

$$F = -kx$$

Frictionless

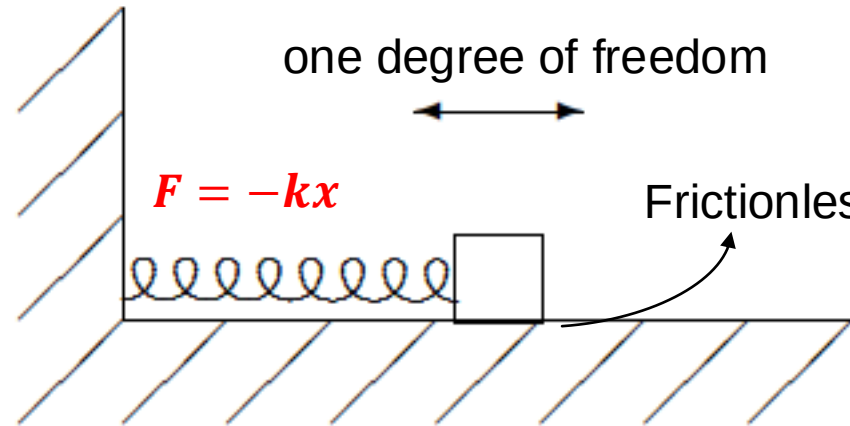
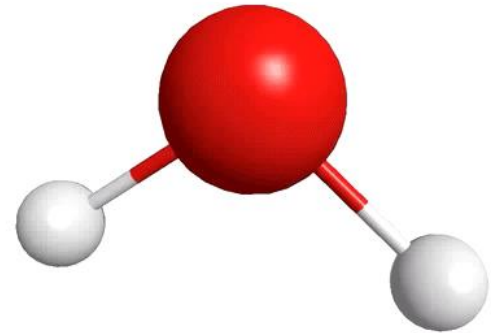
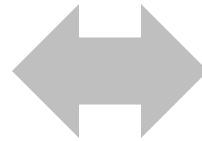
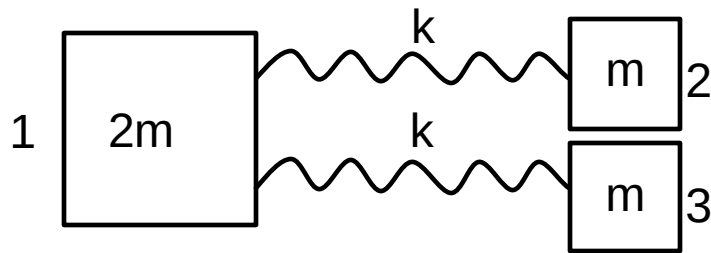


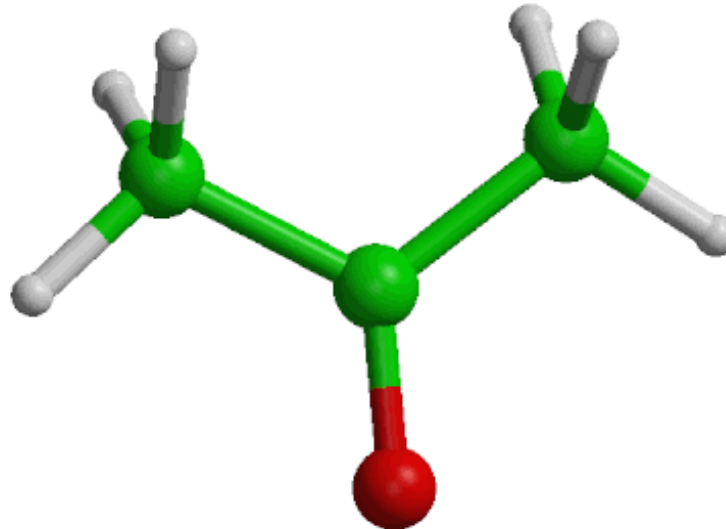
Figure 1.1: A mass on a spring.

$$F = m\ddot{x}$$

2.3 MOLECULAR VIBRATION



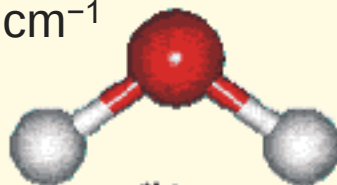
Polyatomic molecules



2.3 MOLECULAR VIBRATION

H₂O Normal modes:
3N-3

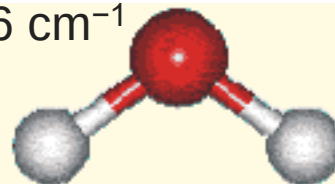
3657 cm⁻¹



ν_1

symmetric stretch

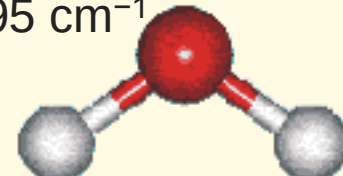
3756 cm⁻¹



ν_3

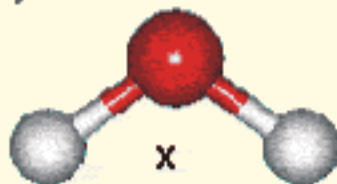
asymmetric stretch

1595 cm⁻¹

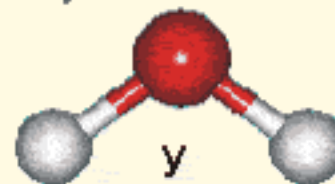


ν_2

bend

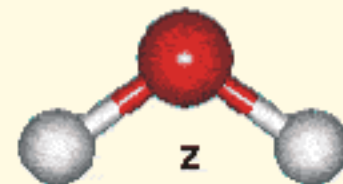


x



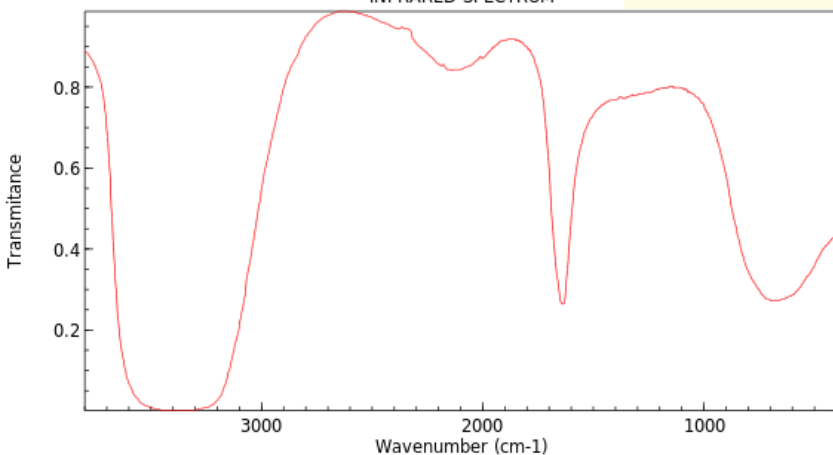
y

librations



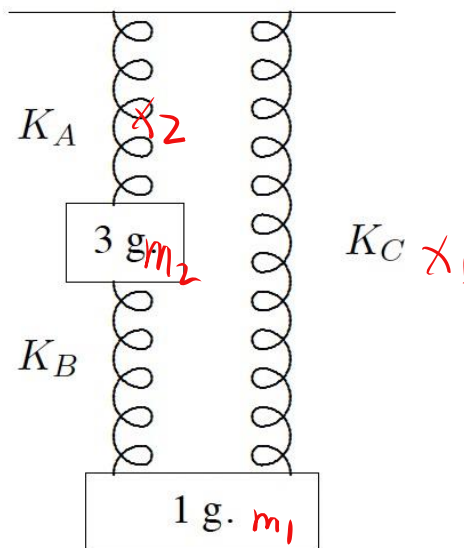
z

WATER
INFRARED SPECTRUM



HOMOWORK

3.2. Consider the vertical oscillation of the system of springs and masses shown below with the spring constants $K_A = 78$, $K_B = 15$ and $K_C = 6$ (all dynes/cm). Find the normal modes, normal coordinates and associated angular frequencies. If the 1 g. block is displaced up 1 cm from its equilibrium position with the 3 g block held at its equilibrium position and both blocks released from rest, describe the subsequent motion of both blocks.



$$\text{for } m_1: \quad -k_c x_1 - k_B(x_1 - x_2) = m_1 \ddot{x}_1$$

$$\text{for } m_2: \quad -k_B x_2 + k_B(x_1 - x_2) = m_2 \ddot{x}_2$$

$$\Rightarrow \begin{bmatrix} -\frac{k_B + k_C}{m_1} & \frac{k_B}{m_1} \\ \frac{k_B}{m_2} & -\frac{k_B + k_C}{m_2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{bmatrix}$$

$$-AX = \ddot{X}$$

$$\det(A - \omega_d^2 I) = 0$$

$$\begin{vmatrix} 21 - \omega_d^2 & -15 \\ -5 & 31 - \omega_d^2 \end{vmatrix} = 0$$

$$\omega_1 = 6 \quad \omega_2 = 4$$

$$\omega = 6$$

$$\begin{pmatrix} -15 & -15 \\ -5 & -5 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 0$$

1 +

$$a_1 = -a_2$$

$$x_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{i(6t+\phi)}$$

$$\omega = 4$$

$$\begin{pmatrix} 5 & -15 \\ -5 & 15 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 0$$

$$a_1 = 3a_2$$

$$x_2 = \begin{pmatrix} 3 \\ 1 \end{pmatrix} e^{i(4t+\phi)}$$

$$t=0 \quad x_1=0 \quad x_2=0 \quad \phi=\rho \Rightarrow$$

$$\therefore x_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \cos(6t) \quad x_2 = \begin{pmatrix} 3 \\ -1 \end{pmatrix} \cos(4t)$$

$$t=0 \text{ 时 } x_1 = -1 \quad x_2 = 0$$

$$\begin{cases} \alpha + 3\beta = -1 \\ -\alpha - \beta = 0 \end{cases} \Rightarrow \alpha = \frac{1}{2} \quad \beta = -\frac{1}{2}$$

$$x_1 = \frac{1}{2} \cos(6t) - \frac{3}{2} \cos(4t)$$

$$x_2 = -\frac{1}{2} \cos(6t) + \frac{1}{2} \cos(4t)$$